INTRODUCTION

This module builds on the prior ones that provided an overview of asset-backed securities (ABS) and described the benefits of securitization, the securitization process, and typical securitization structures. This module focuses on the largest ABS market in the world, mortgage-backed securities (MBS). It introduces mortgage loans and their characteristic features; discusses residential MBS (RMBS), including mortgage pass-through securities and collateralized mortgage obligations (CMOs); and describes commercial MBS (CMBS). It shows how to measure, mitigate, and share securitization-related risks across different tranches and their characteristics, and it examines MBS cash flows and risks.

Category

Mortgage-Backed Security (MBS) Instrument and Market Features

Learning Module Overview

- Prepayment risk is the risk that the borrower does not repay the principal (or a portion of the principal) as scheduled. Contraction risk occurs when the borrower repays the principal faster than anticipated, while extension risk occurs when the borrower repays the principal more slowly than planned.
- Mortgage-backed securities (MBS) are bonds created from securitizing mortgage loans. Mortgage loans provide borrowers the funds to purchase property and require borrowers to repay lenders on a mutually agreed schedule, otherwise the lender has the right to seize the property.
- MBS can be created based on either residential or commercial mortgages. Bonds created from the securitization of mortgages backed by residential properties are residential mortgage-backed securities (RMBS).
- Because both scheduled principal repayments and unscheduled prepayments are made over the life of an MBS, the contractual maturity for an MBS does not reveal its actual payments and prepayments. The weighted average life is a measure widely used to assess when an MBS can be expected to be paid off.
- A mortgage pass-through security is a security created when mortgage lenders pool mortgages together and use them to back securities that they sell to investors. The cash flows of a mortgage pass-through security depend on the monthly cash flows of the underlying pool of mortgages.
- Collateralized mortgage obligations (CMOs) securitize mortgage pass-through securities or multiple pools of loans and are structured to redistribute the cash flows to different bond classes or tranches, thereby creating securities that have different exposures to prepayment risk.
- The tranching structure of a CMO can redistribute prepayment risk across the different tranches; the more senior a tranche is, the less exposure it has to prepayment risk and default risk.
- Commercial mortgage-backed securities (CMBS) can consist of just a few underlying commercial mortgages, and so one default in a CMBS pool may significantly impact the CMBS investors. Investors must evaluate this unique concentration risk by analyzing the individual loans and properties backing the CMBS, the owners of the commercial properties themselves, as well as the CMBS structure.
- Unlike RMBS, CMBS offer investors call protection at either the structural or individual loan level and thus trade more like corporate bonds than RMBS. However,

Self-Assessment

1. Which of the following statements regarding the loan-to-value ratio (LTV) is *most* accurate?

A. The LTV for a given mortgage remains static for the life of the loan.
B. The higher the LTV, the more protection a lender has in the event of default.
C. LTV serves as a key measure in both residential and commercial mortgages.

Hide Solution

Solution:

The correct answer is C. LTV serves as a key measure both in residential and commercial mortgages. LTV is calculated as the ratio of the amount of the mortgage to the property's value. A is incorrect because over time, the LTV changes: As the borrower makes mortgage payments, including principal repayments, the outstanding balance on the loan is reduced and also, fluctuations in the market value of the property cause the borrower's equity to change. B is incorrect because the higher the LTV, the less protection a lender has in the event of default.

2. Which of the following statements regarding mortgage-backed securities (MBS) is *most accurate*? MBS:

A. must be backed or guaranteed by a government or a quasi-government entity.
B. can be created by securitizing mortgages backed by residential or by commercial properties.
C. that use credit enhancements to reduce credit risk for securities backed by residential mortgages typically are agency RMBS.

Hide Solution

Solution:

The correct answer is B. The mortgages in MBS can either be residential mortgages or commercial mortgages. A is incorrect because MBS do not have to be guaranteed by the government or a quasi-government entity. But, when these mortgages or the securities issued backed by these mortgages are guaranteed by the government, there is a credit support. MBS backed by residential mortgages that are issued by private entities and not guaranteed by a federal agency or a government-sponsored enterprise (GSE) are called "non-agency RMBS." C is incorrect because non-agency RMBS, not agency RMBS, typically use credit enhancements, such as insurance, letters of credit, guarantees, or subordinated interests, to mitigate the credit risk and improve the overall quality of the mortgage pool. These securities include credit enhancements because they are not guaranteed or insured by a government agency or by a GSE.

3. When a mortgage is used as collateral for a mortgage pass-through security, the mortgage is *most likely* said to be:

A. enhanced.
B. securitized.
C. subordinated.

Hide Solution

Solution:

incorrect because a mortgage that is used as collateral for a mortgage pass-through security is said to be *securitized*, not enhanced or subordinated.

4. Which of the following structures is created through time tranching?

A. Z-tranches
B. Floating-rate tranches
C. Principal-only (PO) securities

Hide Solution

Solution:

The correct answer is A. Z-tranches and residual tranches are created through time tranching, which means they do not pay interest payments until a pre-set date. B is incorrect because floating-rate tranches carry interest rates that are linked to an index or a market reference rate (MRR) and are used to hedge interest rate risk in portfolios. C is incorrect because principal-only (PO) securities pay only the principal repayments from the pool and can be created either from mortgage pass-throughs or as a tranche in a CMO.

5. Which of the following risks *most likely* explains why the maturity of an MBS declines when interest rates decline?

A. Balloon risk
B. Extension risk
C. Contraction risk

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Solution:

The correct answer is C. Contraction risk is the risk that the borrower might pay back the money borrowed more quickly than anticipated, reducing the amount of future payments the investor receives. When interest rates decline, actual prepayments will be higher than forecasted because homeowners will be more likely to refinance their mortgages. A is incorrect because balloon risk occurs when the borrower fails to make the balloon payment at the maturity of a loan with a balloon payment; it is a form of extension risk. B is incorrect because extension risk is the risk that the borrower might pay back the money borrowed more slowly than anticipated, extending rather than shortening the time of repayment and the maturity of the bond.

6. Which of the following statements related to mortgage-backed securities (MBS) is true?

A. The contractual maturity for an MBS accurately predicts future payments and prepayments.
B. A measure widely used by market participants for MBS is the weighted average life, or simply the average life, of the MBS.
C. A 30-year, option-free corporate bond and an MBS with a 30-year legal maturity with the same coupon rate offer equivalent interest rate risk.

Hide Solution

Solution:

The correct answer is B. A measure widely used by market participants for MBS is the weighted average life, or simply the average life, of the MBS. This measure gives investors an indication of how long they can expect to hold the MBS before it is paid off, assuming interest rates stay at current levels and thus expected prepayments are realized. A is incorrect because the contractual maturity for an MBS does *not* reveal its actual payments and prepayments. Unscheduled principal repayments will shorten the maturity of an MBS, but the pattern of repayments is driven by shifts in interest rates that are not known with

legal maturity with the same coupon rate and thus would not offer equivalent interest rate risk.

7. A major difference between residential and commercial mortgage-backed securities is that:

A. CMBS securitize conforming mortgages and are collateralized by homogeneous properties as compared to properties backing an RMBS.
B. investors must evaluate the individual loans and properties backing a CMBS as well as the owners of the commercial properties themselves.
C. a CMBS pool may consist of thousands of mortgages from various parts of a country, while an RMBS may consist of a few homogenous residential mortgages.

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Solution:

The correct answer is B. CMBS can consist of a few underlying commercial mortgages, whereas an RMBS pool may consist of thousands of mortgages from various parts of a country. Thus, one default in a CMBS pool may have a significant impact on the CMBS investors. Investors in the CMBS must consider explicitly this unique concentration risk by not only analyzing the CMBS structure but also the individual loans and properties backing the CMBS as well as the owners of the commercial properties themselves. A is incorrect because RMBS, not CMBS, securitize conforming mortgages and are collateralized by homogeneous (single-family) residential properties. C is incorrect because an RMBS pool, not a CMBS pool, may consist of thousands of mortgages from various parts of a country, while CMBS can consist of a few underlying commercial mortgages.

8. Which of the following statements related to the debt service coverage (DSC) ratio, a key indicator of potential credit performance in commercial real estate lending, is *most accurate*? A DSC ratio:

A. is equal to the property's annual net operating income (NOI) divided by its debt service.
B. is calculated as the ratio of the amount of the loan to the underlying property's value.
C. that exceeds 1.0 indicates that the cash flows from the property are insufficient to cover the debt service while maintaining the property in its initial state of repair.

Hide Solution

Solution:

The correct answer is A. The DSC ratio is equal to the property's annual NOI divided by its debt service. NOI is defined as the rental income reduced by cash operating expenses and a replacement reserve. Debt service is the annual amount of interest payments and principal repayments. B is incorrect because the ratio of the amount of the loan to the underlying property's value is the LTV ratio, another key indicator of potential credit performance, not the DSC. C is incorrect because a DSC ratio that exceeds 1.0 indicates that the cash flows from the property are sufficient, not insufficient, to cover the debt service while maintaining the property in its initial state of repair.

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